Sahaj Bhadja

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Summary

Software Engineer with 2+ years building ML and data infrastructure in **Python**: **PyTorch** model training pipelines over 100K+ logs, **Spark**/Databricks and ETL pipelines, production LLM serving with fallbacks, and **AWS** cloud systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Data Retrieval, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant async **model inference** orchestration with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ LLM jobs at 3x throughput without cascading failures under concurrent load.
- Built **data pipelines** across PostgreSQL, SQL Server, and cloud storage with incremental loads, validation, and reconciliation, running 5K+ monthly jobs at 99%+ reliability and cutting query latency 71%.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Served multimodal models in production behind a switchable **inference layer** routing across Qwen-VL, GPT-4o Vision, and Claude Vision with timeout fallback, handling 10K+ daily requests on AWS.
- Built concurrent batch processing with ThreadPoolExecutor over 1K+ files per run and per-call token and cost **observability** logged to MySQL on AWS RDS, sustaining 90%+ grading consistency for 150+ instructors.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Trained a GPT-based Decision Transformer in **PyTorch** for offline RL on features from 100K+ trajectory logs and HD-map vectors, with Hydra configs and W&B tracking, improving long-horizon planning 18%.
- Built an end-to-end **feature pipeline** aligning ego state, 5-second 10Hz trajectory history, and nearby agents to a global frame, then benchmarked encoders in closed-loop simulation, improving planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL** on AWS Glue and S3 that transformed raw JSON, CSV, and Parquet across raw, intermediate, and final storage tiers, boosting pipeline efficiency 68% and cutting storage costs 15%.
- Optimized large-scale API integrations with indexing, caching, and pagination, improving response times 30%, and built object-oriented Singleton and Factory modules with 20+ automated **test** suites in Jenkins CI/CD.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built hourly **Spark** micro-batch pipelines on Databricks ingesting 50K+ booking and payment events, aggregating peak booking hours, most-used services, daily active users, and revenue for internal dashboards.
- Developed Python Flask APIs with Firebase **real-time** sync and transactional locking against double-booked time slots, sustaining sub-200ms latency and 40% faster booking confirmations for a home-services marketplace.

Skills

Languages: Python, Java, C++, SQL, TypeScript, Shell
ML & Modeling: PyTorch, TensorFlow, Transformers, Feature Engineering, Offline RL, Random Forest, LSTM, Hydra, Weights & Biases
Data Infrastructure: Apache Spark, Databricks, AWS Glue, ETL Pipelines, Incremental Loads, PostgreSQL, SQL Server, MySQL
Serving & Systems: Model Inference, FastAPI, REST APIs, Async Job Orchestration, Concurrency, Distributed Systems, Caching
Cloud & DevOps: AWS (S3, RDS, Glue, ECS/Fargate), Docker, Kubernetes, CI/CD, Jenkins, Linux, Monitoring
Fundamentals: Data Structures, Algorithms, OOP, Design Patterns

Projects

Real-Time Feature Store and Prediction Service | *Python, Spark, Redis, gRPC, PyTorch, MLflow, Kubernetes*

- Designed a feature store where Spark jobs materialize batch features into **Redis** for a gRPC **prediction service** that caches MLflow-versioned PyTorch and LightGBM models in memory and logs predictions for drift monitoring.

CaptionCraft Live | *PyTorch, Transformers, BLIP, GIT, ViT-GPT-2, Flask*

- Fine-tuned and benchmarked BLIP, GIT, and ViT-GPT-2 against a custom model on 10K images and 100K captions, raising accuracy from 81% to 96%, and deployed real-time **inference APIs**.

Trend Tracker with Fashion Recommendation System Live | *CNN, React.js, Strapi, AWS EC2*

- Built and deployed a CNN-based **recommendation** model on AWS EC2 for an e-commerce trend tracker, improving recommendation accuracy from 78% to 92% and lifting engagement across 3M+ users.